Chris Dai

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Education

Harvard University — Aug. 2026 – Expected May 2028
M.S. Health Data Science, T.H. Chan School of Public Health — *Boston, Massachusetts, USA*

University of Michigan, Ann Arbor — Aug. 2024 – May 2026
B.S.E. Industrial and Operations Engineering; Minor in Computer Science — *Ann Arbor, Michigan, USA*

UM–SJTU Joint Institute, Shanghai Jiao Tong University — Sep. 2022 – Aug. 2026
B.S.E. Electrical and Computer Engineering — *Shanghai, China*

Dean's Honor List (2024, 2025) · Myun Lee Scholarship (2025)

Skills

LLM & Evaluation: SFT corpus construction, synthetic dialogue generation & data flywheels, LLM-as-judge evaluation, RAG & embedding retrieval (bge-m3), prompt architecture, agent benchmarking

ML & Data: PyTorch, K-means / PCA, time-series feature engineering, SQL, PostgreSQL, dbt, evaluation-harness design

Engineering: Python, TypeScript, JavaScript, FastAPI, Flask, React, Node.js, Next.js, Playwright / Chrome DevTools Protocol

Experience

AISpeech — Jun. 2026 – Aug. 2026
Sponsored Capstone Thesis — LLM data flywheel for in-vehicle voice assistants — *Shanghai, China*

- Led the data flywheel end to end for the multimedia domain of a 5-person voice-assistant project: built a neuro-symbolic user simulator — **41** persona archetypes over an agenda state machine — emitting retrieval-grounded rollouts against a bge-m3 tool index.
- Reward-scored every rollout on the team's 397B model on a GPU cluster, keeping only provably correct (reward=1) traces, then regenerated 200 dialogues for each of the 11 highest-error functions; spurious no-tool calls fell **4,424→1,410**.
- Produced **17,200** verified (reward=1) multimedia fine-tuning dialogues — 15,000 base plus 2,200 error-targeted — as a four-quadrant corpus (positive/negative × single/multi-turn), each batch cleared by the same automated quality gate.
- Fine-tuned the assistant's model on this corpus, lifting tool-call accuracy **72.8%→82.7%** on real conversations, **76.8%→98.3%** on the synthetic validated set and **83.4%→96.0%** on the user-profile set — graded by an independent DeepSeek-V4-Pro judge, not the 397B corpus scorer.

ByteDance — May 2026 – Aug. 2026
AI Solutions Intern; Stack: Python, Flask, React, TypeScript, Node.js, Playwright / CDP — *Shanghai, China*

- Built an **LLM-agent evaluation platform** for two enterprise desktop agents: designed a suite of **200+** real user tasks, drove both Electron clients over the Chrome DevTools Protocol, normalized their incompatible telemetry into one trace schema, and scored them with an LLM-as-judge harness on output quality, token throughput and cost — one comparable read, and the team's basis for selection.
- Productized customer technical due diligence into a **citation-grounded RAG** research tool on internal AI tooling, where every conclusion resolves to its source; adopted by the regional client team as their working tool.
- Distilled the team's enterprise deep-research workflow into reusable team-wide skills: one research pass, **2 h → 5 min**.
- Hand-coded all **11** agents' skills for a client-facing demo site — industry-specific and general-purpose — including a PPTX-to-HTML skill the team still uses daily, plus **13** case studies on a shared asset base of **240+** industry reference materials.

Shenzhen Legion Technology Co., Ltd. — Apr. 2025 – Apr. 2026
Data Engineering & AI Intern (part-time); Stack: Python, FastAPI, PostgreSQL, React/TS — *Remote, China*

- Co-developed **MetaFlow** (two-person project, started Mar. 2026), a metadata-centric NL-to-SQL system that answers questions through a verifiable relational IR instead of generating SQL directly; I owned the metadata layer, evaluation suite and studio front end — my collaborator owned the query engine.
- Built its gold-case regression suite — **20** scenarios pinned across 5 pipeline stages as contract-as-test — plus a contracts endpoint running data-quality rules as executable checks, so a schema change is caught by re-running the suite, not by hand.
- Earlier in the role, designed the prompt architecture of an open-source RAG data-governance copilot for dbt pipelines — question classification (catalog / SQL / data-QA), entity extraction, embedding-based semantic retrieval — and its evaluation framework of fixed test cases for retrieval accuracy and governance compliance, owning feature scope and release priorities, not the upstream codebase.

PGF Technology Group — Mar. 2026 – Apr. 2026
Consulting Engagement (4-person team), sole developer; Stack: Next.js, TypeScript, Postgres — *Rochester Hills, Michigan, USA*

- Sole developer of the software on a four-person consulting engagement: shipped a full-stack work-order tracking system covering all **9** production stations of a ∼50-person harness manufacturer — a complete solution the client took into internal use after handover.
- Diagnosed stale reads across serverless instances and migrated storage in production (in-memory → Blob → Postgres); pinned the scan, queue and alert flows with **6** Playwright e2e suites.

Projects

Cardiac Rehabilitation Wearable-Data Study — Apr. 2025 – Jan. 2026
SURE Research Program, University of Michigan — Advisors: Profs. Mariel Lavieri & Jessica Golbus — *Ann Arbor, Michigan, USA*

- Built a time-series feature pipeline over **400,000+** raw step records from **247** enrolled cardiac-rehab patients — rolling means, trend slopes, periodicity, plus imputation for gapped and high-variance series.
- Phenotyped recovery trajectories with K-means/PCA and detected plateaus with a Kalman filter: against the standard 36-session protocol, ∼**68%** of the 191-patient analysis cohort was discharged off its true plateau.

Real-Time EMG Biofeedback & Muscle-Synergy Analysis — Oct. 2024 – Apr. 2026
NeuRRo Lab, University of Michigan — PI: Chandramouli Krishnan — *Ann Arbor, Michigan, USA*

- Built the lab's real-time EMG biofeedback trainer and muscle-synergy pipeline (MATLAB, Unity/C#), both now standard tools in ongoing clinical experiments; ported its multi-objective gradient-descent optimizer from **PyTorch** to Unity C# for on-device adaptation.